# Simplifying radical expressions

**ANSWER KEY**

## Core Concepts: Simplifying radical expressions

**1** For the model $f(x) = 3x + 6$ in simplifying radical expressions, evaluate $f(2)$.

**a.** 12 ✓

**b.** 15

**c.** 6

**d.** 16

Substitute $x = 2$: $f(2) = 3(2) + 6 = 12$.

**2** Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

**a.** $x = 1$ and $x = 7$

**b.** $x = 1$ **and** $x = 6$ ✓

**c.** $x = -1$ and $x = -6$

**d.** $x = 7$ and $x = 6$

Factoring $(x - 1)(x - 6) = 0$ yields the roots $x = 1$ and $x = 6$.

**3** A data set recorded for simplifying radical expressions consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

**a.** 18.5

**b.** 19.5

**c.** 20.5 ✓

**d.** 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives $20.5$.

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## Procedural Fluency & Algebraic Manipulation

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**4** For the model $f(x) = 3x + 6$ in simplifying radical expressions, evaluate $f(3)$.

**a.** $15$ ✓

**b.** $18$

**c.** $9$

**d.** $19$

Substitute $x = 3$: $f(3) = 3(3) + 6 = 15$.

**5** Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

**a.** $x = 2$ and $x = 8$

**b.** $x = 2$ **and** $x = 7$ ✓

**c.** $x = -2$ and $x = -7$

**d.** $x = 9$ and $x = 14$

Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

**6** A data set recorded for simplifying radical expressions consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

**a.** $18.5$

**b.** $19.5$

**c.** $20.5$ ✓

**d.** $22.5$

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by $4$ gives $20.5$.

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## Graphical & Geometric Interpretation

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**7** For the model $f(x) = 3x + 6$ in simplifying radical expressions, evaluate $f(4)$.

**a.** $18$ ✓

**b.** $21$

**c.** $12$

**d.** $22$

Substitute $x = 4$: $f(4) = 3(4) + 6 = 18$.

**8** Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

**a.** $x = 3$ and $x = 9$

**b.** $x = 3$ **and** $x = 8$ ✓

**c.** $x = -3$ and $x = -8$

**d.** $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

**9** A data set recorded for simplifying radical expressions consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

**a.** $18.5$

**b.** $19.5$

**c.** $20.5$ ✓

**d.** $22.5$

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by $4$ gives $20.5$.

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## Exam-Style Problem Solving & Modeling

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**10** For the model $f(x) = 3x + 6$ in simplifying radical expressions, evaluate $f(5)$.

**a.** $21$ ✓

**b.** $24$

**c.** $15$

**d.** $25$

Substitute $x = 5$: $f(5) = 3(5) + 6 = 21$.

**11** Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

**a.** $x = 4$ and $x = 10$

**b.** $x = 4$ **and** $x = 9$ ✓

**c.** $x = -4$ and $x = -9$

**d.** $x = 13$ and $x = 36$

Factoring $(x - 4)(x - 9) = 0$ yields the roots $x = 4$ and $x = 9$.

**12** A data set recorded for simplifying radical expressions consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

**a.** $18.5$

**b.** $19.5$

**c.** $20.5$ ✓

**d.** $22.5$

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by $4$ gives $20.5$.

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